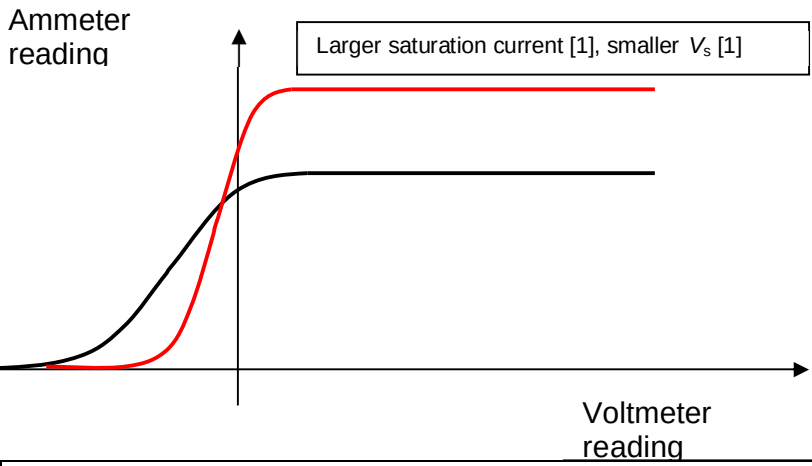


			p.d across the appliance is $V = IR = 10.0 \times 0.1 = 1.0 \text{ V}$	
		(ii)	p.d across appliance = $20 / 100 \times 5.0 = 1.0 \text{ V}$ current = $1/10.0 = 0.10 \text{ A}$	1
		(iii)	$\varepsilon = 45.0 / 100 \times 5.0 = 2.25 \text{ V}$	1

6	(a)	The emission of electrons from the surface of a metal when EM radiation [1] of high enough frequency [1] is incident on the surface.		
	(b)	(i)	 Ammeter reading Larger saturation current [1], smaller V_s [1] Voltmeter reading	
	(b)	(ii)	1. Increasing the intensity while keeping frequency constant increases the number of photons arriving per unit time. [1] Thus, the number of electrons emitted per unit time increases. Hence current increases proportionately. [1]	
	(b)	(ii)	2. Photons travel at the speed of light. As soon as a photon collides with an electron, its energy is immediately (and fully) imparted to it. [1]	
	(c)	(i)	1. Using gain in KE = loss in EPE, we have $\frac{1}{2} m v^2 = e V$ $\Rightarrow v = \sqrt{(2eV / m)}$ $= \sqrt{(2e \times 625 / 9.11 \times 10^{-31})}$ $= 1.4817 \times 10^7$ [1] Since $\lambda_{dB} = h / (mv)$ we have $\lambda_{dB} = 6.63 \times 10^{-34} / (9.11 \times 10^{-31} \times 1.4817 \times 10^7) = 4.9117 \times 10^{-11} \text{ m}$ [1]	
	(c)	(i)	2. From the given formula, $d = 1 \times 4.9117 \times 10^{-11} \div (2 \times \sin 9.96^\circ)$ [1] $= 1.42 \times 10^{-10} \text{ m}$ [1]	
	(c)	(ii)	Rings become larger. [1] P.d. decreases $\rightarrow$ KE / speed of incident electrons on film decreases [1]	

			→ de Broglie wavelength increases [1] → diffraction angle, θ increases [1]	
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